The empirical recurrence for OEIS A235749, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

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Abstract

OEIS A235749 counts arrays of width 2 over $\{0, 1, 2, 3\}$ by reducing every 2×2 subblock to a single number—here the sum of the two middle entries—and requiring the resulting derived array to have its rows lexicographically nonincreasing and its columns lexicographically nondecreasing. The entry carries an empirical recurrence of order 32 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The row comparison is settled by three consecutive rows of the array; the column comparison is not settled by any bounded window, but what is still outstanding about it is one bit per adjacent pair of columns, so the count is a walk count on $S = 101$ states.

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1 The conjecture

OEIS A235749 is "Number of (n+1)X(1+1) 0..3 arrays with the sum of each 2X2 subblock two median terms lexicographically nondecreasing columnwise and nonincreasing rowwise.". It has offset 1 and begins

$$256, 2648, 26852, 231182, 1922672, 14817810, 110573053, 791130664, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 16*a(n-1) -3*a(n-2) -1116*a(n-3) +3195*a(n-4) +36776*a(n-5) -144355*a(n-6) -775876*a(n-7) +3409369*a(n-8) +11994752*a(n-9) -52035329*a(n-10) -145821540*a(n-11) +551539761*a(n-12) +1428073416*a(n-13) -4145223065*a(n-14) -11151370108*a(n-15) +21729158076*a(n-16) +67484344064*a(n-17) -73656117248*a(n-18) -305148542976*a(n-19) +117869487680*a(n-20) +984593256704*a(n-21) +184038815744*a(n-22) -2112780488704*a(n-23) -1459803293696*a(n-24) +2607031529472*a(n-25) +3342204194816*a(n-26) -1030546145280*a(n-27) -3447562665984*a(n-28) -1135057305600*a(n-29) +1114979696640*a(n-30) +934281216000*a(n-31) +203843174400*a(n-32)

As of the "Last modified" line on the live entry (Jul 23 2025 08:46:28, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The arrays have 2 columns and a growing number of rows, with entries in $\{0, 1, 2, 3\}$. A 2×2 subblock of such an array is

$$B = \begin{pmatrix} p & q \\ r & s \end{pmatrix}, \qquad p = x(i, j),\ q = x(i, j + 1),\ r = x(i + 1, j),\ s = x(i + 1, j + 1).$$

Write $v_1 \le v_2 \le v_3 \le v_4$ for the four entries of B sorted into order. The entry attaches to B the single number

$$f(B) \;=\; v_2 + v_3,$$

the sum of the two middle entries. An array with R rows and 2 columns has $(R-1)\times 1$ subblocks, and they form a derived array

$$D(i,j) = f\left(\begin{pmatrix} x(i,j) & x(i,j+1) \\ x(i+1,j) & x(i+1,j+1) \end{pmatrix}\right), \qquad 0 \le i < R-1,\;\; 0 \le j < 1.$$

Write $\preceq$ for the lexicographic order on finite sequences of integers: $u \preceq w$ when $u = w$, or when $u_k < w_k$ at the first index k where they differ. The condition is that the rows of D, read left to right and compared in this order, are nonincreasing from top to bottom,

$$D_i \;\succeq\; D_{i+1} \qquad \text{for every } i,$$

and that the columns of D, read top to bottom and compared in the same order, are nondecreasing from left to right,

$$D^j \;\preceq\; D^{j+1} \qquad \text{for every } j.$$

3 A bounded state

Grow the array one row at a time. A row of D is built from two consecutive rows of the array, so the row condition is decided three array rows at a time and needs no memory beyond the derived row just formed. The column condition is different: which of two columns of D is the smaller is decided by the first derived row in which they differ, and that row can be arbitrarily far down. What is bounded is not the comparison but what remains open about it.

Lemma 1. *Fix j and suppose the derived rows $D_0, \ldots, D_i$ have been written. If $D_t(j) = D_t(j+1)$ for every $t \le i$, then $D^j \;\preceq\; D^{j+1}$ holds for the finished array if and only if, at the first later row t with $D_t(j) \ne D_t(j+1)$, the inequality $D_t(j) < D_t(j+1)$ holds; and if no such row occurs the two columns are equal and the condition holds. If instead $D_t(j) \ne D_t(j+1)$ for some $t \le i$, then the comparison is already settled by the earliest such t and no later row can change it.*

Proof. Immediate from the definition of the lexicographic order: it looks only at the first index at which the two sequences differ, and prefixes that agree contribute nothing. □

So a single bit per adjacent pair of columns—"this pair has agreed in every derived row so far"—carries everything the future needs to know, and there are 0 such pairs. The state is therefore

$$\big(\text{the array row just written, the derived row just formed, the 0 bits}\big),$$

with a distinguished start standing for the empty array, so that a walk of n steps is an array of n rows. A step appends a row, forms the new derived row, checks it against the previous one for the row condition, and for each pair still flagged either leaves the flag set (the pair agreed again) or clears it after checking the required inequality once. Every state may end an array, so $\tau = \mathbf{1}$ and

$$a(n) \;=\; \iota^{\top} M^{\,j} \tau, \qquad S = 101.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 2. *Let $q(t) = t^{32} - \sum_i c_i t^{32-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^j q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+32} - \sum_i c_i M^{j+32-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 3.

$$a(n) = 16\,a(n-1) - 3\,a(n-2) - 1116\,a(n-3) + 3195\,a(n-4) + 36776\,a(n-5) - 144355\,a(n-6) - 775876\,a(n-7) + 340$$

for every $n > 30$.

Proof. The vector $w = q(M)\tau$ was formed by 32 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 30 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array of width 2 over $\{0, 1, 2, 3\}$ with two and with three rows was written out cell by cell, the derived array formed, and its rows and its columns compared as whole words. No state, no bits of Lemma 1 and no recurrence are involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 2 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A235749.

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